



Department of Education

STEM Education
National Schools of Excellence
STEM Physics Syllabus
Grade 11
Term 2

<u>DOMAIN</u>	<u>Phylum</u>
SCIENCE	Physics
Technology	
Engineering	
Mathematics	

Issued free to schools by the Department of Education

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Acknowledgements

This STEM Physics Syllabus for Grade 11 constitutes ten (10) Thematic Areas (Thema) and is intended for students enrolling for STEM Education. In this booklet only three Thema are presented.

Ultima Thulë Ltd was engaged by the Department of Education to develop the STEM Curriculum. The content was generated by writers from universities, secondary schools, new university graduates with knowledge in the subject matter and industry subject experts. Quality assurance groups and the Science Subject Advisory Committee have also contributed to the development of this syllabus.

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SECRETARY'S MESSAGE

This Physics Syllabus is to be used exclusively by teachers of Physics to teach Grade 11 Students enrolling for the STEM Education through the Schools of Excellence throughout Papua New Guinea. The STEM Physics Bridging Syllabus ensured that Grade 11 students enrolling for STEM Education were grounded in the basic concepts espoused in Physics before undertaking the advanced concepts introduced in this syllabus.

The syllabus conforms to the visions of the National Education Plan and the School of Excellence Policy. Furthermore, it is undergirded by a policy and legislative matrix that were issued by the National Government over the last decades, such as the Vision 2050 and Development Strategic Plan 2030. The Preamble of the National Constitution, especially the First Directive of the *National Goals and Directive Principles* makes a clarion call for the Integral Human Development, which the Department took heed of to ensure, through this new syllabus, that *all forms of beneficial creativity, including sciences and cultures, [are] to be actively encouraged*. A principle object of the *National Education Act 1983 (amended 1995)* reiterates this virtue of Integral Human Development (Section 4).

In accordance with the widespread demand for a home-grown curriculum to replace the existing one with the view to position the country as a bastion of innovation in the new century the Secretary of Education has embarked on introducing the STEM Curriculum to the Schools of Excellence. It is hoped that students coming through the STEM Education will not only gain productive employment in the knowledge-based economy but become creators, innovators and inventors of knowledge and ultimately commercialize their creativity through development of their intellectual property rights.

In addition, there are four core elements espoused in this syllabus, insofar as student development is concerned: Students are expected to gain knowledge and skills and apply them to solve every day problems whilst appreciating Physics as a fundamental field of science.

This STEM Physics Syllabus for Grade 11 constitutes ten (10) Thematic Area (Thema) and is intended for students enrolling for STEM Education. In this booklet only three (3) Thema are presented. By studying these Thema students are expected to gain the foundational knowledge needed in Physics.

The success of this new Curriculum is highly dependent on the Teachers. They play a pivotal role by being innovative and keeping abreast of new information on scientific research and innovations. The challenge for teachers of Physics is to engage students to learn realistic context for increased and better understanding and appreciate the relevance of Physics to human lives, including good health and environment.

In order to ensure Teachers discharge their responsibilities adequately this syllabus is accompanied by a Teacher's Guide and Teacher's Resource Book which provides sufficient teaching and assessment materials to help Teachers.

I commend and approve this syllabus as the official curriculum for Physics to be used for the Grade 11 STEM Education in the Schools of Excellence throughout Papua New Guinea.

DR UKE KOMBRA, PHD
Secretary for Education

INTRODUCTION

Physics is the study of matter and its composition, how they interact to form new compounds through chemical reactions and the energy they generate as a result of these interactions. The concepts taught in Physics help us to understand these interactions at the molecular and atomic levels. Every substance, whether naturally occurring or artificially produced, consists of one or more elements of the periodic table.

Since Physics is an experimental science, laboratory work becomes an essential part of the syllabus. Practical work can be used to enable students to investigate the properties and reactions of substances and to provide opportunities for students to learn and test the principles and concepts of Physics. Since Physics is a highly specialized subject students undertaking Chemistry are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language skills.

The Grade 11 Physics Syllabus entails ten (10) Thematic Areas (Thema) and these are listed below. Each Thema has a single Module and three topics. Over time different Modules will be introduced. The modules in this Syllabus are sequenced to guide teachers of Physics through the process of imparting basic chemistry knowledge and skills to students.

THEMA	MODULE	TOPICS
BIODIVERSITY	Ethnobotany	1.Medicinal/Herbal Plants 2.Pharmacology of Traditional Plants 3.Therapeutic Drugs
BIOTECHNOLOGY	Principles of Biotechnology	1.Polymerase Chain Reaction 2.Dye Chemistry in Gene Sequencing 3.Proteins
HEALTH AND MEDICINE	Biochemistry in Medicine	1.Drug Design 2.Pharmacology 3.Significance of Diagnostic Laboratory Tests
EARTH SCIENCE		
WASTE MANAGEMENT		
INFRASTRUCTURE		
ENERGY		
BATTERY TECHNOLOGY		
INFORMATION SYSTEM		
COMPUTER SCIENCE		
ARTIFICIAL INTELLIGENCE		
ENTERPRENEURSHIP		

RATIONALE

There is a confluence of events that necessitated the need to introduce STEM Curriculum to the Schools of Excellence:

1. The School of Excellence Policy that was launched by Prime Minister Honorable James Marape, MP, in June 2020, calls for the establishment of the Schools of Excellence and the introduction of STEM Curriculum;
2. The Government has decided to replace the Observation Based Education to Standards Based Education in 2019 and tasked the Department of Education to introduce the STEM Curriculum as part of its reform agenda;
3. There is public outcry, both from public and private sectors, over the quality of graduates coming through the National Education System; and
4. The changing landscape in the technology domain globally and the pressing need to produce the next generation of workforce that is agile and can adopt to the changing circumstances as well as become enterprising individuals through their creation of intellectual properties.

The combination of these factors has led the Department of Education to adopt the STEM Curriculum and to introduce this to the Schools of Excellence as a necessary disruptive intervention in an effort to develop the next generation of national intellectual capital and to drive the growth and prosperity of the country through creation of intellectual properties.

In developing the STEM Curriculum, it was realized that the Grades 9 and 10 Science Syllabi does not provide a firm foundation to launch students into STEM Curriculum at the Schools of Excellence. Consequently, a STEM Bridging Program was necessary. The STEM Bridging Syllabus is a bridge across the divide that Grade 11 Students enrolling for STEM Education must cross.

In the Physics Bridging Syllabus students were introduced to the basic concepts and knowledge of Chemistry in order for them to develop an appreciation of the discipline in a broader context. In this syllabus students are introduced to advanced concepts in Physics as they are applied to the ten Thema.

The introduction of the STEM Curriculum, especially the Bridging Program and the Grades 11 and 12 Syllabi, will cause consequential changes to the Upper Classes of STEM Education and the Lower Secondary Syllabi. These syllabi will be introduced in the course of time.

NEW NOMENCLATURE

As part of the overall Curriculum Development new nomenclature were introduced. These are explained below:

Term	Explanation
Domain	Each STEM area is defined as <i>domain</i> (e.g. Science Domain, Technology Domain, etc.)
Phylum	What would normally be regarded as “subject” (e.g. Biology, Physics, Chemistry, etc.) is now regarded as <i>phylum</i> ; the plural is <i>phyla</i> .
Thema	Thematic areas of studies
Module	A specific concept within the Thema which is being studied
Topic	A specific aspect of the Module that is being studied

STEM Domain

At this juncture of our national development there is no other task before us more important than deploying a robust STEM Curriculum that develops the next generation of highly educated students to enter the work force. By building on the best of current practices and standards the STEM Curriculum aims to take us, beyond the constraints of the existing structures of schooling, towards a shared vision of excellence nationally and marketing the nation globally in terms of Science, Technology, Engineering and Mathematics. In this STEM Curriculum, there are ten (10) Thema (Thematic Areas) introduced.

Science

It is the study of the natural world, including the laws of the nature associated with physics, chemistry and biology and the treatment or application of facts, principles, concepts or conventions associated with these disciplines. Science is both a body of knowledge that has been accumulated over time and a process-scientific inquiry-that generates new knowledge. Knowing from science informs the engineering design process. In this STEM Curriculum the Science Domain has three Phyla – Biology, Chemistry and Physics.

Technology

It comprises the entire system of people and organizations, knowledge, processes, and devices that go into creating and operating technological artifacts, as well as the artifacts themselves. Throughout history humans have created technology to satisfy their wants and needs. Much of modern technology is the product of science and engineering and technological tools used in both fields. There are two Phyla – Information Technology and Computer Sciences – introduced in this STEM Curriculum for the Technology Domain.

Engineering

It concerns both the knowledge about the design and creation of products and a process for solving problems. The design process must result in a final product that must operate or function under various constraints. One constraint in engineering design is the laws of nature. Engineers must learn to harness the natural forces at work around them and through their design process develop products that must operate in harmony with nature. Other constraints include things such as time, money, available materials, ergonomics, environmental regulations and manufacturability. Engineering utilizes concepts in science and mathematics as well as technological tools. There are three (3) Phyla introduced in the Engineering Domain – Civil, Mechanical and Electrical.

Mathematics

It is the study of patterns and relationships among quantities, numbers, and shapes. Specific branches of mathematics include arithmetic, geometry, algebra, trigonometry and calculus. Mathematics is used in science and in engineering. In this STEM Curriculum, Mathematics is considered as a Service Domain; specific topics tailored towards specific Thema are also considered relevant to undergird the relevance and applicability of Mathematics Domain to other fields of knowledge.

STEM PHYSICS CONTENT OVERVIEW

Table 1.1: Thema Sequence and Duration

Grade 11 Term 2: STEM-proper Course		
BIODIVERSITY (3 Weeks)	1. Measuring Biodiversity and Waves	1. Electromagnetic Waves in Vegetation Monitoring 2. Electromagnetic Waves in Forest Mapping 3. Sound Waves in Identifying Animals
BIOTECHNOLOGY (3 Weeks)	1. Measuring Biomolecules through Electricity and Magnetism	1. DNA-Transistor 2. Magnetic Resonance 3. Force Spectroscopy
HEALTH AND MEDICINE (3 Weeks)	1. Medical Physics	1. Ultrasound 2. X-ray 3. CT-Scan

Table 1.2 Sequencing & Benchmarking STEM Physics Content Standards

Setting STEM Physics Content Standards for Progressive Learning & Assessment National STEM Assessment			
Domain	Science	Module Content Standard	
Phylum	Physics	11.1: Discuss and analyze the importance of Biodiversity's measurements and its application in solving problems in the real world.	
Module 11.1	Measuring Biodiversity and Waves		
Topics	Performance Indicators	Module Benchmark	Assessment Tasks
1. Electromagnetic Waves in Vegetation Monitoring	11.1.1a: Define Waves and Electromagnetic Waves 11.1.1b Differentiate between Absorption, Transmission and Reflectance behavior of Wave 11.1.1c: Identify the Electromagnetic Spectrum 11.1.1d: Define Radiation 11.1.1e: Classify the different types of Radiation 11.1.1f: Describe how Radiation Interacts with Matter 11.1.1g: Explain how satellite sensors works. 11.1.1h Define Spectral signature of objects 11.1.1i Discuss how RGB images work 11.1.1j Explain how health monitoring of vegetation works	Student's ability to master knowledge and skills of; - Waves - Electromagnetic Wave Principles - Radiation Principles - Satellite Imaging - Vegetation Mapping - Forest Mapping - Sound Waves - Sound Conversion Process - Animal Identification Process	- Assignment - Module Test - Lab Work - End of Term Exam
2. Electromagnetic Waves in Forest	11.1.2a: Understand that Forest mapping process is similar to		

Mapping	Vegetation Monitoring 11.1.2b: Explain how Forest Mapping process works			
3. Sound Waves in Identifying Animals	11.1.3a: Define Sound Waves 11.1.3b: Understand Sound Production Mechanism 11.1.3c: Explain the process of Sound Conversion: Analog to Digital 11.1.3d: Discuss and describe how animals sound are captured and recorded.			
Module 11.2	Measuring Biomolecules Through Electricity and Magnetism	Module Content Standard		
		11.2a: Discuss and describe the process of measuring biomolecules 11.2b: Describe and analyze DNA-Transistor, Magnetic Resonance and Force Spectroscopy 11.2c: Define Electricity and Magnetism		
Topics	Performance Indicators		Module Benchmark	Assessment Tasks
1. DNA-Transistor	11.2.1a: Define and describe the Electric Principles forming the DNA-transistor 11.2.1b: Define DNA-Transistor 11.2.1c: Understand the process of DNA Sequencing using DNA-Transistor		Student's ability to master knowledge and skills of; - Principles of Electricity - Principles of Magnetism - DNA-Transistor - Magnetic Resonance - Force Spectroscopy - Measurement - Biomolecule Types - DNA Sequencing - Force Analysis	- Assignment - Module Test - Lab Work - End of Term Exam
2. Magnetic Resonance	11.2.2a: Describe and analyse Magnetic Resonance 11.2.2b Define Magnetism Principles used in Magnetic Resonance 11.2.2c Recognize importance of Magnetic Resonance in Diagnosis and Treatment			
3. Force Spectroscopy	11.2.3a: Define Mechanical Forces 11.2.3b: Define Binding Forces 11.2.3c: Define Force Spectroscopy and describe its process 11.2.3d: Apply Force Spectroscopy Techniques to measure biomolecules			
Module 11.3	Medical Physics	Module Content Standard		
		11.3.a: Explain medical diagnosis using ultrasound 11.3.b: Explain medical diagnosis and analysis using X-ray 11.3.c: Explain medical diagnosis and analysis using		

		Computerized Tomography Scan	
Topics	Performance Indicators	Module Benchmark	Assessment Tasks
1. Ultrasound	11.3.1a: Define sound and differentiate ultrasound 11.3.1b: Explain ultrasound in the Echolocation Process 11.3.1c: Apply Echolocation to identify fetus in the womb	Student's ability to master knowledge and skills of; - Principles of Medical Diagnosis and Treatment - Application of Sound Waves - Application of X-Ray waves - Application of Computerized Tomography Scan.	- Assignment - Module Test - Lab Work - End of Term Exam
2. X-Ray	11.3.2a: Define and Describe X-ray Waves 11.3.2b: Discuss How X-ray is used in Medical Diagnosis 11.3.2c: State some of the application of X-ray techniques		
3. Computerized Tomography Scan	11.3.3a: Define Computerized Tomography 11.3.3b: Discuss how Computerized Tomography Works 11.3.3c: Explain the process of medical diagnosis using Computerized Tomography Scan		

Table 1.3 Mapping Content Progressions by Modules

Thema/Module	Broad Content Standard	11.1	11.1	11.1
Biodiversity/ 11.1 Measuring Biodiversity and Waves	11.1: Discuss and analyze the importance of Biodiversity's measurements and its application in solving problems in the real world.	✓		
Biotechnology/ 11.1 Measuring Biomolecules through Electricity and Magnetism	11.2a: Discuss and describe the process of measuring biomolecules 11.2b: Describe and analyze DNA-Transistor, Magnetic Resonance and Force Spectroscopy 11.2c: Define Electricity and Magnetism		✓	
Health and Medicine/ 11.1 Medical Physics	11.3.a: Explain medical diagnosis using ultrasound 11.3.b: Explain medical diagnosis and analysis using X-ray 11.3.c: Explain medical diagnosis and analysis using Computerized Tomography Scan			✓

THEMA BIODIVERSITY

MODULE 11.1: MEASURING BIODIVERSITY AND WAVES

3weeks

In this module we will see how Physics is used in the aid of measuring Biodiversity. Physics is the study of the Laws of Nature and those Laws are applicable to Biodiversity in many ways. Measuring Biodiversity will help Biologists, Environmental scientists and other dedicated scientists who want those measurements to make better informed decision. In addition, it will broaden their knowledge in understanding the distribution of Biodiversity across the planet. This module will enable the student to grasp the real-life skill of measuring and interpreting data based on physical laws.

Broad Content Standard

11.1 Discuss and analyze the importance of Biodiversity's measurements and its application in solving problems in the real world.

Content Benchmark

Student's ability to master knowledge and skills of;

- Waves
- Electromagnetic Wave Principles
- Radiation Principles
- Satellite Imaging
- Vegetation Mapping
- Forest Mapping
- Sound Waves
- Sound Conversion Process
- Animal Identification Process

Topic 1: Electromagnetic Waves in Vegetation Monitoring

Performance Indicators

Students are expected to:

- 11.1.1a: Define Waves and Electromagnetic Waves
- 11.1.1b: Differentiate between Absorption, Transmission and Reflectance behavior of Wave
- 11.1.1c: Identify the Electromagnetic Spectrum
- 11.1.1d: Define Radiation
- 11.1.1e: Classify the different types of Radiation
- 11.1.1f: Describe how Radiation Interacts with Matter
- 11.1.1g: Explain how satellite sensors works.
- 11.1.1h: Define Spectral signature of objects
- 11.1.1i: Discuss how RGB images work
- 11.1.1j: Explain how health monitoring of vegetation works

Content Expansion

Body of Knowledge for their Understanding

- Wave Theory
 - Structure of a Wave
 - Amplitude
 - Wavelength
 - Velocity
 - Classification of Waves
 - Transverse
 - Longitudinal
 - Mechanical
 - Matter
 - Electromagnetic
- Electromagnetic Wave
- Electromagnetic Wave Behavior
 - Absorption
 - Transmission
 - Reflection
 - Scatter
 - Diffraction
 - Refraction
- Electromagnetic Spectrum
 - Radio Wave
 - Microwave
 - Infrared
 - Visible Light
 - Ultraviolet
 - X-ray
 - Gamma
 - Classification
 - Alpha
 - Beta
 - Gamma and X-ray Electromagnetic Radiation
- Radiation
 - Electromagnetic

- Mechanical
- Nuclear
- Cosmic
- Remote Sensing
 - Satellite
 - Satellite Sensors
 - Resolution of Satellite Sensors
 - Spectral Signature
 - Spectral Imaging
 - Health monitoring Of Vegetation
 - Photosynthesis
 - Near-Infrared Wavelengths
 - Normalized Difference Vegetation Index

Set of Skills for their proficiency levels

- Intermediate understanding of Waves and Remote Sensing
- Experience in knowing cross-discipline studies
- Having the ability to experience the knowledge of Remote Sensing utilized in Real World

Trait of Attitudes and Values to govern their lives

- Systematic mindset in understanding how different ideas are related to form a working system.
- Appreciation of Physics Waves Concept being applied to Biodiversity

Topic 2: Electromagnetic Waves in Forest Mapping

Performance Indicators

Students are expected to;

11.1.2a: Define Forest Mapping

11.1.2b: Identify the electromagnetic waves used in Forest Mapping

11.1.2c: Differentiate between Forest Mapping and Vegetation Monitoring

11.1.2d: Explain how Forest Mapping process works

Content Expansion

Body of Knowledge for their Understanding

- Forest Mapping
- Remote Sensing Platforms and Sensors
- Data Interpretation of Forest Mapping

Set of Skills for their proficiency levels

- Ability to understand mapping and survey contexts
- Experience another application of Remote Sensing
- Comprehension of Data Interpretation

Trait of Attitudes and Values to govern their lives

- Creativity in applying the same Electromagnetic Wave Principle to different applications.
- Willingness to explore technology and science.
- Focus on inventions and innovations.

Topic 3: Sound Waves in Identifying Animals

Performance Indicators

Students are expected to:

11.1.3a: Define Sound Waves

11.1.3b: Understand Sound Production Mechanism

11.1.3c: Explain the process of Sound Conversion: Analog to Digital

11.1.3d: Discuss and describe how animals sound are captured and recorded.

Content Expansion

Body of Knowledge for their Understanding

- Sound Wave Principles
 - Sound Definition
 - Sound Behavior
 - Frequency
 - Wavelength
 - Amplitude
 - Speed of Sound
 - Sound Reflection/Echoes
- Electronic Analog To Digital Conversion
 - Analog-to-Digital Converter
 - Sampling and Rate
 - Bit Resolution
- Automatic Sound Recognition System
 - Sound Capturing
 - Sound Matching and Identification

Set of Skills for their proficiency levels

- Understanding of the fundamental of Sound Waves
- Experience in Conversion of Electric Signal to Digital Signal
- Ability to comprehend Sound Conversion process as applied to Animal Sound

Trait of Attitudes and Values to govern their lives

- Appreciation of the relationship between sound waves and electromagnetic waves
- Ability to apply the principles of sound to other applicable areas.

Application of Knowledge, Skills and Values (AKSA)

1. Discuss significance of knowledge gained in real time situations
 - The skills of Remote Sensing
 - Application Of Electromagnetic Waves
 - Awareness of importance of Technology
2. Discuss significance of knowledge gained in real time situations
 - The skills of Forest Mapping and Monitoring.
 - Application of Electromagnetic Waves and Remote Sensing Skills.
 - Effectiveness of Theory and Application in Real Life.
3. Discuss significance of knowledge gained in real time situations
 - Understanding of Converting natural phenomena into digital simulation
 - Application of Analog to Digital Conversion
 - Comprehension of Animal Sound Recognition System

THEMA BIOTECHNOLOGY

Module 11.2: Measuring Biomolecules Through Electricity and Magnetism

3weeks

Biomolecules are critical elements in all living things. They come in different sizes at a microscopic and nanoscopic levels at the cellular level. By analyzing their physical size, this could provide vital insights into their structure and function. Biomolecules are classed into four main groups known as Carbohydrates, Lipids, Nucleic-acid and Proteins. In this module, we'll take a look at the different physical techniques used in analyzing Biomolecules. This module will be more basic in explaining concepts in a systematic manner without emphasizing on the calculations.

Broad Content Standard

11.1 Discuss and analyze the importance of Biotechnological devices and its application in understanding Biomolecules.

Content Benchmark

Student's ability to master knowledge and skills of;

- Principles of Electricity
- Principles of Magnetism
- DNA-Transistor
- Magnetic Resonance
- Force Spectroscopy
- DNA Sequencing
- Force Analysis

Topic 1: DNA-Transistor

Performance Indicators

Students are expected to;

11.2.1a: Define DNA-Transistor

11.2.1b: Define and describe the Electric Principles applied in the DNA-transistor

11.2.1c: Understand the process of DNA Sequencing using DNA-Transistor

Content Expansion

Body of Knowledge for their Understanding

- DNA Sequencing
 - Biomolecules
 - DNA
 - DNA Sequencing
- Electronic Transistor Device
 - Electricity
 - Electric Current
 - Current Flow
 - Voltage
 - Potential Difference
 - Resistance
 - Ohm's Law
 - Electric Field
 - Transistor
- DNA-Transistor technique
 - Architecture of DNA Transistor

Set of Skills for their proficiency levels

- Ability to integrate Electricity and DNA concepts
- Fundamental awareness of problem solving in a different area of knowledge

Trait of Attitudes and Values to govern their lives

- Appreciation of the relationship between Electricity and DNA Structure
- Ability to apply the principles of DNA Transistor to DNA Sequencing.

Topic 2: Magnetic Resonance Spectroscopy

Performance Indicators

Students are expected to;

11.2.2a: Describe and analyse Magnetic Resonance

11.2.2b: Define Magnetism Principles used in Magnetic Resonance

11.2.2c: Recognize importance of Magnetic Resonance in Diagnosis and Treatment

Content Expansion

Body of Knowledge for their Understanding

- Principles of Magnetism
 - Magnetism
 - Magnetic field and Field Lines
- Magnetic Resonance Spectroscopy
 - Magnetic Resonance
 - Spectrometer
 - Fourier Transform
 - Chemical Shift

Set of Skills for their proficiency levels

- Ability to understand magnetism
- Recognize magnetic spectroscopy resonance as derived from magnetism
- Understanding of how magnetic field interacts with biochemical compounds.
- Comprehension of Biochemical information extracted from magnetic resonance spectroscopy

Trait of Attitudes and Values to govern their lives

- Acknowledging the Scientific principles in medical fields
- Acquisition of Problem Solving mindset in Biotechnology
- Critical thinking Skills in understanding the mechanism of Magnetic Resonance Spectroscopy

Topic 3: Force Spectroscopy

Performance Indicators

Students are expected to;

11.2.3a: Define Mechanical Forces

11.2.3b: Define Binding Forces

11.2.3c: Define Force Spectroscopy and describe its process

11.2.3d: Apply Force Spectroscopy Techniques to measure biomolecules

Content Expansion

Body of Knowledge for their Understanding

- Atomic Force Microscopy Principles
- Atomic Force Microscopy Imaging
- Force-Distance Curve
- Operation Modes
 - Contact Modes
 - Dynamic (Tapping) Mode
 - Non-contact Mode

Set of Skills for their proficiency levels

- Recognize that force is a fundamental and natural phenomena
- Understand that biomolecules have forces that characterizes their structure
- Comprehension of Atomic Force Microscopy as a Technology for measuring forces within the biochemical compounds of nature.

Trait of Attitudes and Values to govern their lives

- Appreciate that atomic structures constitutes macro structures
- Ability to use technology as a tool to diagnose problems

Application of Knowledge, Skills and Values (AKSA)

1. Discuss significance of knowledge gained in real time situations
 - Application of Simple Electric Principles
 - Skills of DNA Separation
 - Understanding of DNA structure
2. Discuss significance of knowledge gained in real time situations
 - The skills of Magnetic Resonance Spectroscopy
 - Appreciation of Magnetism
 - Understanding of Magnetic Spectroscopy application in measurement
 - Application of Magnetic Resonance in Medical Field
3. Discuss significance of knowledge gained in real time situations
 - Understanding of Force as a natural process
 - Skill of Force Spectroscopy Analysis
 - Measurement Technique using Force Spectroscopy Analysis

THEMA HEALTH AND MEDICINE

MODULE 11.3: MEDICAL PHYSICS

3 weeks

Medical Physics is a study that basically utilizes the concept of Radiation in medical analysis to help diagnose patients. One of its main functions is to analyse the internal structures of the human body through imaging techniques. Medical physics is also used as treatment, prevention and diagnostic measures for diseases. It is very critical in hospitals and healthcare industries. By understanding medical physics, health workers may have a very handy tool in correctly analyzing sick patients.

Broad Content Standard

11.1 Discuss and analyze the importance of Medical Physics and its application in medical diagnosis and treatment.

Content Benchmark

Student's ability to master knowledge and skills of;

- Principles of Medical Diagnosis and Treatment
- Application of Sound Waves
- Application of X-Ray waves
- Application of Computerized Tomography Scan

Topic 1: Ultrasound

Performance Indicators

Students are expected to;

11.3.1a: Define sound and differentiate ultrasound

11.3.1b: Explain ultrasound in the Echolocation Process

11.3.1c: Apply Echolocation to identify fetus in the womb

Content Expansion

Body of Knowledge for their Understanding

- Sound and Ultrasound Wave Principles
 - Sound
 - Frequency
 - Speed
 - Classification of Sound Frequency
 - Infrasound
 - Audible Range
 - Ultrasound
- Echolocation
 - Concept
- Ultrasound Imaging in Fetus Diagnosis
 - Ultrasound Machine
 - Probes
 - Scanner Unit
 - Display unit

Set of Skills for their proficiency levels

- Comprehension of Ultrasound as a type of sound wave
- Ability to relate ultrasound in biotechnology
- Understanding of Ultrasound imaging
- Experience of Fetus Identification Process

Trait of Attitudes and Values to govern their lives

- Ability to relate innovation in Physics and Medical Fields
- Awareness of understanding Scientific Principles and application simultaneously
- Fosters self-reliable learning through motivated learning
- Focus on problem solving ideas

Topic 2: X-Ray

Performance Indicators

Students are expected to;

11.3.2a: Define and Describe X-ray Waves

11.3.2b: Discuss How X-ray is used in Medical Diagnosis

11.3.2c: State some of the application of X-ray techniques

Content Expansion

Body of Knowledge for their Understanding

- Principles of X-Ray Waves/Radiation
 - Concept
 - Hard and Soft Rays
 - X-Ray interaction with matter
 - Sources of X-ray
- X-Ray Machine
 - Production of X-ray
- Radiology
 - X-ray procedure in Medical Diagnosis
 - X-ray image
 - Attenuation
 - X-ray Uses

Set of Skills for their proficiency levels

- Ability to comprehend X-ray's
- Understand X-ray production
- Grasp the x-ray machine in X-ray production
- Understand X-ray as a medical tool

Trait of Attitudes and Values to govern their lives

- Possessing of an analytical mindset towards understanding processes
- Integrated knowledge for innovation
- Appreciation towards science in health and medicine

Topic 3: Computerized Tomography Scan

Performance Indicators

Students are expected to;

11.3.3a: Define Computerized Tomography

11.3.3b: Discuss how Computerized Tomography Works

11.3.3c: Explain the process of medical diagnosis using Computerized Tomography Scan

11.3.2c: State some of the application of CT Techniques

Content Expansion

Body of Knowledge for their Understanding

- Computerized Tomography Scanner
- Methods
 - Volume CT Technology
 - Slip-Ring Technology
- CT Imaging
 - Detection
 - Pitch of Imaging

Set of Skills for their proficiency levels

- Recognizing CT Scan as a 3-D X-ray
- Ability to comprehend CT technologies
- Understanding CT mechanism
- Relating understanding of CT in Medical Imaging

Trait of Attitudes and Values to govern their lives

- Ability to view situations in multiple perspectives
- Ability to apply scientific thinking to solve everyday problems
- Appreciation of the use of CT scan in Medical diagnosis

Application of Knowledge, Skills and Values (AKSA)

1. Discuss significance of knowledge gained in real time situations
 - Understanding of Ultrasound
 - Knowledge of Echolocation Mechanism
 - Comprehension of Obstetric Process
2. Discuss significance of knowledge gained in real time situations
 - Knowledge of X-Ray
 - Skill of X-Ray in Radiology/Radiography
3. Discuss significance of knowledge gained in real time situations
 - Knowledge of Computerized Tomography
 - Skills of Computerized Tomography in Medical Application

REFERENCES

1. <https://www.physicsclassroom.com/Class/waves/u10l1b.cfm>
2. <https://byjus.com/physics/waves/>
3. <https://www.sciencelearn.org.nz/resources/121-behaviour-of-waves>.
4. https://en.wikipedia.org/wiki/Electromagnetic_spectrum
5. <https://www.sciencedirect.com/science/article/pii/S0741521405012838>
6. <https://www.usgs.gov/fags/what-remote-sensing-and-what-it-used>
7. <http://www.edc.uri.edu/nrs/classes/NRS409/RS/Lectures/HowRemoteSensorWork.pdf>
8. https://en.wikipedia.org/wiki/Spectral_signature
9. <https://www.intechopen.com/books/geoscience-and-remote-sensing/remote-sensing-of-forest-health>
10. <https://www.nasa.gov/specials/X59/science-of-sound.html>
11. https://www.researchgate.net/publication/320162958_Animal_Recognition_System_Based_on_Convolutional_Neural_Network
12. <https://www.geeksforgeeks.org/analog-to-digital-conversion/>
13. <https://www.ibm.com/ibm/history/ibm100/us/en/icons/dnatransistor/>
14. <https://www.hilarispublisher.com/open-access-journals/dna-sequencing-7692.html>
15. <https://radiologykey.com/obstetric-and-gynecologic-imaging/>
16. <https://www.healthline.com/health/ct-scan>
17. <https://science.howstuffworks.com/x-ray.htm>
18. <https://www.nibib.nih.gov/science-education/science-topics/magnetic-resonance-imaging-mri>
19. <https://www.britannica.com/science/echolocation>
20. <https://www.sciencedirect.com/book/9780323445481/obstetric-imaging-fetal-diagnosis-and-care>
21. <https://www.nibib.nih.gov/science-education/science-topics/computed-tomography-ct>
22. <https://www.fda.gov/radiation-emitting-products/medical-imaging/ultrasound-imaging>

